

CLASS 10 / OCT 27 / U07

DEADLINE WEEK + VOLUMETRIC DATA

EVERYTHING THROUGH ARRAYS IS IN · PROJECT 4 BEGINS

GSM + P2b + P3c DUE

MID-SEMESTER DEADLINE / P4a ISSUED TODAY / P3d CHECKPOINT TODAY

DUE TODAY — THE MID-SEMESTER DEADLINE

Everything through Arrays must be in today. This is a regular class after that.

GSM

GRAPHIC STANDARDS MANUAL

DUE WEEK 10 — Oct 27 / 7%

Four pages, each rule beside a tested application. Revised again for the booklet.

P2b

CSG 3D PRINTED OBJECT

DUE WEEK 10 — Oct 27 / 7%

The print and its process sheet, handed in, with the source, mesh and slicer files.

P3c

ATTRACTOR: 2D FIELD DRAWING

DUE WEEK 10 — Oct 27 / 7%

One 17 × 11 sheet: line drawing, grayscale field and legend. SVG, PDF and graph.

Already due and expected in: P2a, P3a, P3b.

P3d CHECKPOINT TODAY · BRING YOUR SITE'S POINT CLOUD AND AT LEAST TWO MAPPED LAYERS WITH LEGENDS.

P3d is due next class, Week 11 — Nov 3. P4a is issued today.

A MESH OR A FIELD

Two ways to describe the same solid. Until now you have only used one.

MESH — THE SKIN

vertices + edges + faces

Stores only the surface. Inside and outside are implied by the faces.

A Boolean has to cut and rebuild faces, and can fail.

FIELD — A VALUE EVERYWHERE

value = $f(x, y, z)$

Every point in space has a number. The surface is where the number crosses a threshold.

A Boolean is arithmetic on two numbers. It cannot fail.

YOU HAVE ALREADY DRAWN A FIELD: THE P3c GRAYSCALE. THIS IS THE SAME IDEA, IN 3D.

DENSITY OR SIGNED DISTANCE

Two kinds of field. P4a is built on the second.

DENSITY

How much is here.

Counted from samples, such as points within a radius. High inside a cluster, zero far away.

The surface is a threshold you choose.

SIGNED DISTANCE

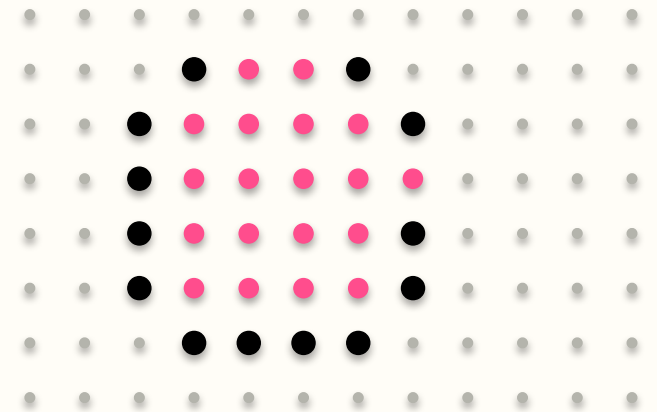
How far to the surface, with a sign.

$d < 0$ inside

$d = 0$ the boundary

$d > 0$ outside

The surface is always at zero.



● inside ● zero

● outside — a circle, sampled

INSIDE, OUTSIDE AND THE ZERO BOUNDARY: SAY WHICH ONE YOU ARE LOOKING AT.

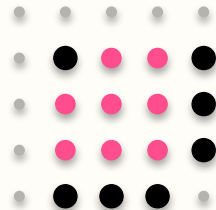
A SHAPE IS A FUNCTION

Give it a point, get back a signed distance. No vertices anywhere.

SPHERE

```
sphere(p, c, r):  
    return length(p - c) - r
```

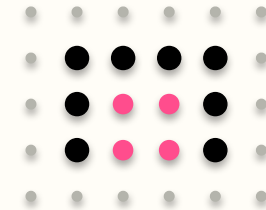
Distance to the centre, minus the radius. On the surface that is exactly zero.



BOX

```
box(p, c, b):  
    q = abs(p - c) - b  
    return length(max(q, 0))  
           + min(max(q.x, q.y, q.z), 0)
```

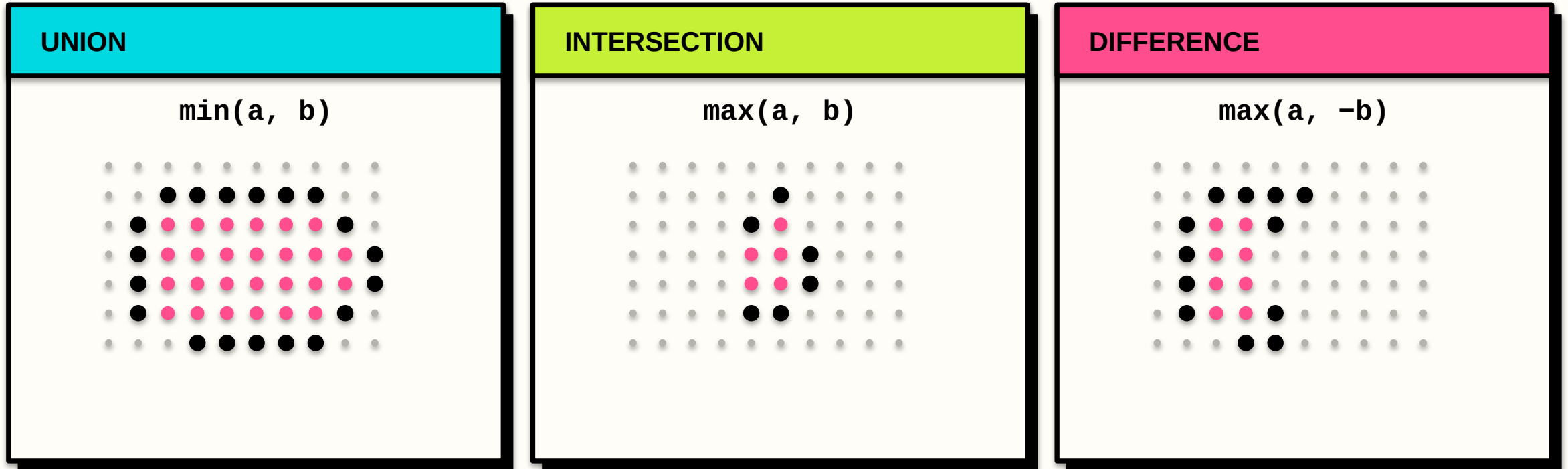
b is the half-size on each axis.
Same idea, one more line.



CHANGE r OR b AND THE WHOLE FIELD CHANGES. THE PARAMETER IS STILL LIVE.

THE BOOLEANS AGAIN — AS MIN AND MAX

The same three operations from P2. On fields, each is one line.



UNION KEEPS THE NEARER SURFACE. INTERSECTION KEEPS THE FARTHER. DIFFERENCE FLIPS b INSIDE OUT, THEN INTERSECTS.

Same operands as P2, so the same rule: $a - b$ is not $b - a$.

BLEND, SAMPLE, EXTRACT

Three steps turn a function into something you can see, slice and cut.

SMOOTH BLEND

```
smooth_min(a, b, k)
```

$k = 0$ is a hard union. Larger k fills the joint with a soft fillet. A mesh Boolean cannot do this.

RESOLUTION

```
sample(field, bounds, n)
```

The field is read on a grid. Coarse is fast and faceted. Fine is smooth and slow. State the number.

SURFACE EXTRACTION

```
extract(grid, iso=0.0)
```

A mesh is built where the samples cross zero. Only now does the volume have faces.

P4a ASKS YOU TO STATE THE SAMPLING RESOLUTION AND THE EXTRACTION SETTINGS.

THE SAME THING, WRITTEN OUT

The top half reads a point cloud. Today is the bottom half: a shape as a function.

```
cloud = load("site.ply")
print(len(cloud))                # state this before

cloud = decimate(cloud, keep=0.05)
print(len(cloud))                # and after

for p in cloud:
    p.density = len(within(cloud, p, radius))

volume = voxelize(cloud, size=0.5)

# A designed field can also define a volume.
def field(p):                    # a shape as a function
    a = sphere_sdf(p, center_a, r_a)
    b = box_sdf(p, center_b, size_b)
    return smooth_min(a, b, k)    # the blend comes free

grid    = sample(field, bounds, resolution)
surface = extract(grid, iso=0.0)  # where the field is zero
```

WHAT IT MAPS ONTO

sample	scalar field
point cloud	implicit surface
origin	signed distance field
units	isosurface
decimation	threshold
density	noise
voxel	resolution
radius	

FROM “def field” DOWN IS P4a: TWO FIELDS, ONE BLEND, ONE GRID, ONE SURFACE AT ZERO.

ISSUED TODAY

P4a — SDF VOLUME

DUE WEEK 12 — Nov 10 / 14%

Project 4 begins. Build a volume from signed distance fields — the CSG logic from P2, applied to fields instead of meshes — and choose the one you will slice and laser-cut in P4b.

COMPARE

SAME FIELDS

Union, intersection and difference of the same sources.

VARY

THREE VARIANTS

Three controlled parameter changes on the chosen result.

STATE

HOW IT WAS READ

Resolution, extraction settings, and the spatial intention.

REQUIREMENTS — P4a

Visual quality 35% · computational understanding 25% · technical execution 20% · experimentation 10% · requirements and identity 10%.

01

Use two 17 x 11 inch sheets to explain a volume built from signed distance fields (SDF) — the CSG logic from P2, applied to fields instead of meshes.

02

Compare union, intersection and difference using the same source fields. Show the chosen result and three controlled parameter variants.

03

Diagram the field logic and identify density, signed distance or another implicit function. State sampling resolution and surface extraction settings.

04

Select one volume for P4b and explain the spatial intention. Submit the editable graph and retain the source fields.

CHECKPOINT · WEEK 11 — Nov 3 — Bring an SDF diagram, a union/intersection/difference comparison and a candidate volume for slicing.

NOW — YOUR FIRST FIELD VOLUME

Rest of the class: two primitives, three Booleans, one surface.

✓

A sphere field sampled on a grid and extracted at zero

✓

A box field beside it, each with its size exposed

✓

Union as min, then swapped for max, then for $\max(a, -b)$

✓

A smooth blend, with k changed until you can see the joint

✓

The same volume extracted at two resolutions, and the numbers written down

IF YOU CAN SAY WHERE ZERO IS, YOU CAN SAY WHERE THE SURFACE IS.

BEFORE NEXT CLASS

NEXT

Week 11 — Nov 3: SDF volume studio work — primitives, the three Booleans, smooth blend, noise and surface extraction.

P3d

Due next class, Week 11 — Nov 3: one sheet with each layer small, with its legend, beside the combined view, and the working file.

P4a

Checkpoint Week 11 — Nov 3. Bring an SDF diagram, a union/intersection/difference comparison and a candidate volume for slicing.

DUE

P4a is due Week 12 — Nov 10 (14%): two sheets, the editable graph and the source fields. Later dates are provisional.